AP CALCULUS AB

Unit 1 • Lesson 1-6 • Guided notes

Lesson 1-6 • Average Rate of Change

Guided notes • Fill in the blanks and work the examples in the boxes

Learning goals

- Compute an average rate of change from a formula, a table or a graph.
- Interpret an average rate of change in context with units, including average velocity.
- Use the difference quotient $\frac{f(a+h)-f(a)}{h}$.

Vocabulary. average rate of change, secant line, difference quotient, average velocity, units.

Key idea

The average rate of change of f on $[a,b]$ is _____, the slope of the _____ through $(a, f(a))$ and $(b, f(b))$. On $[a, a+h]$ it is the **difference quotient** _____.

Example 1 • Formula

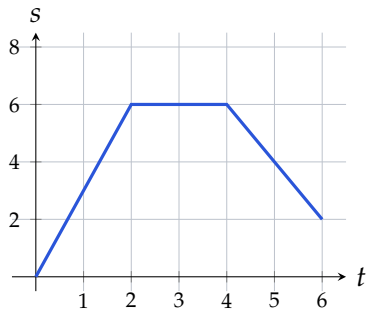
Find the average rate of change of $f(x) = x^3 - 2x$ on $[-1, 2]$, and simplify $\dfrac{f(1+h)-f(1)}{h}$ for $f(x) = x^2$.

Example 2 • Table, with units

t (min)	0	5	10	15	20
$A(t)$ (L)	40	52	61	66	68

Find and interpret the average rate of change of A on $[5, 15]$.

Example 3 • Average velocity from a graph



The position $s(t)$ (meters) of a particle is graphed. Find its average velocity on $[0,2]$, $[2,4]$ and $[0,6]$.

Common mistakes

- Giving the change $f(b) - f(a)$ instead of the rate.
- Forgetting units, or interpreting an average rate as a rate at an instant.

Check your understanding

1. The average rate of change of $g(x) = \frac{12}{x}$ on $[2,6]$ is _____.
2. For $f(x) = x^2 + x$, $\frac{f(2+h)-f(2)}{h} =$ _____.